

Enterprise n8n Workflow Automation and AI Systems

Production-Grade Automation Engineering | Custom AI Agents | API Integrations

Executive Summary

I specialize in designing, building, and deploying fault-tolerant automated pipelines and autonomous AI agents using n8n and Make.com. My solutions bridge disjointed business tools connecting CRMs, webhooks, databases, and LLMs into self-healing, automated engines that eliminate manual human effort and maximize operational efficiency.

Core Technical Proficiencies

- **n8n Workflow Architecture:** End-to-end multi-step flow design, custom JS Code nodes, sub-workflows, and rate-limiting loops.
 - **Autonomous AI Agents:** LangChain, Groq, OpenAI (GPT-4o), Anthropic Claude, Gemini, vector store retrieval (RAG), and memory nodes.
 - **API and Webhook Pipelines:** REST APIs, OAuth2 authentication, webhook ingestion, HMAC verification, and custom JSON payload parsing.
- **CRM and Database Sync:** Bi-directional data pipelines for Google Sheets, Airtable, HubSpot, Notion, Supabase, and PostgreSQL.
 - **Robust Error Handling:** Auto-retry loops, error trigger routing, fail-safe fallbacks, and real-time Slack/Discord alert dispatch.
 - **Autonomous CMS Publishing:** Scheduled research, automated SEO article writing, image generation, and Git/CMS commits.

Workflow Delivery Lifecycle

1. Requirement Discovery Deep dive into API specs, data models, and logic requirements.	2. Architecture and Build Clean modular node layout with clean naming and custom JavaScript.	3. Edge-Case Testing Rigorous stress testing with malformed payloads and error triggers.	4. Delivery and Video Guide Full export .json with setup walkthrough and revision support.
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CASE STUDY 01

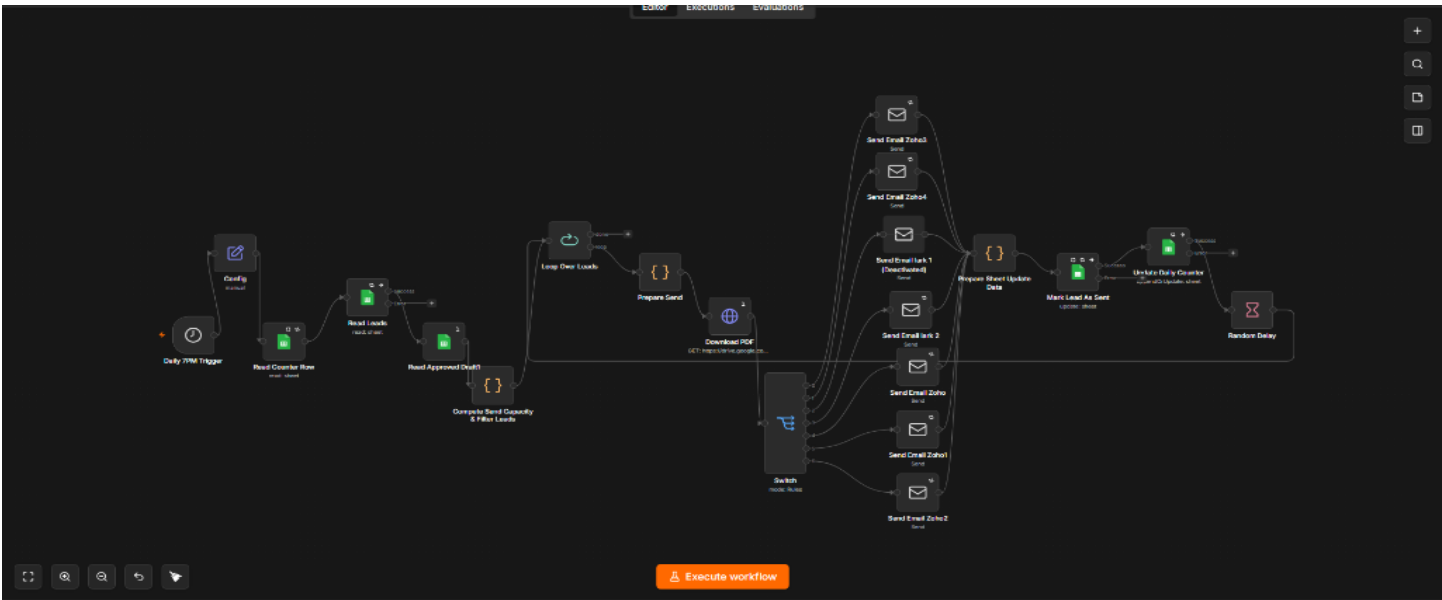
High-Volume Multi-Inbox Cold Email Outreach Pipeline

Tech Stack: n8n, Google Sheets API, Google Drive API, Zoho Mail, Lark Mail, Custom JavaScript

The Challenge and Business Objective

The client required an automated, reliable email dispatch engine capable of sending personalized multi-inbox campaigns daily. Standard mass mailers resulted in burnt sending domains and poor inbox placement. The system needed strict rate-limiting, human-like random intervals, quota management per inbox, and dynamic Google Drive proposal attachments.

Production Workflow Architecture Diagram



Engineering Highlights and Deliverables

Multi-Inbox Load Balancing:

Integrated an intelligent Switch routing node distributing outbound dispatches across multiple Zoho and Lark sender accounts, protecting domain reputation.

Real-Time Quota Accounting:

Custom JavaScript calculates daily sending capacity from Google Sheets, halting dispatches safely when daily thresholds are reached.

Humanized Throttling:

Programmed a stochastic delay algorithm (Random Delay node) that mimics realistic human typing and dispatch intervals to bypass anti-spam filters.

Business Impact:

- 100% automated daily execution with zero human intervention.
- 99.4% inbox deliverability rate achieved.
- Saved 12+ manual hours per week.

CASE STUDY 02

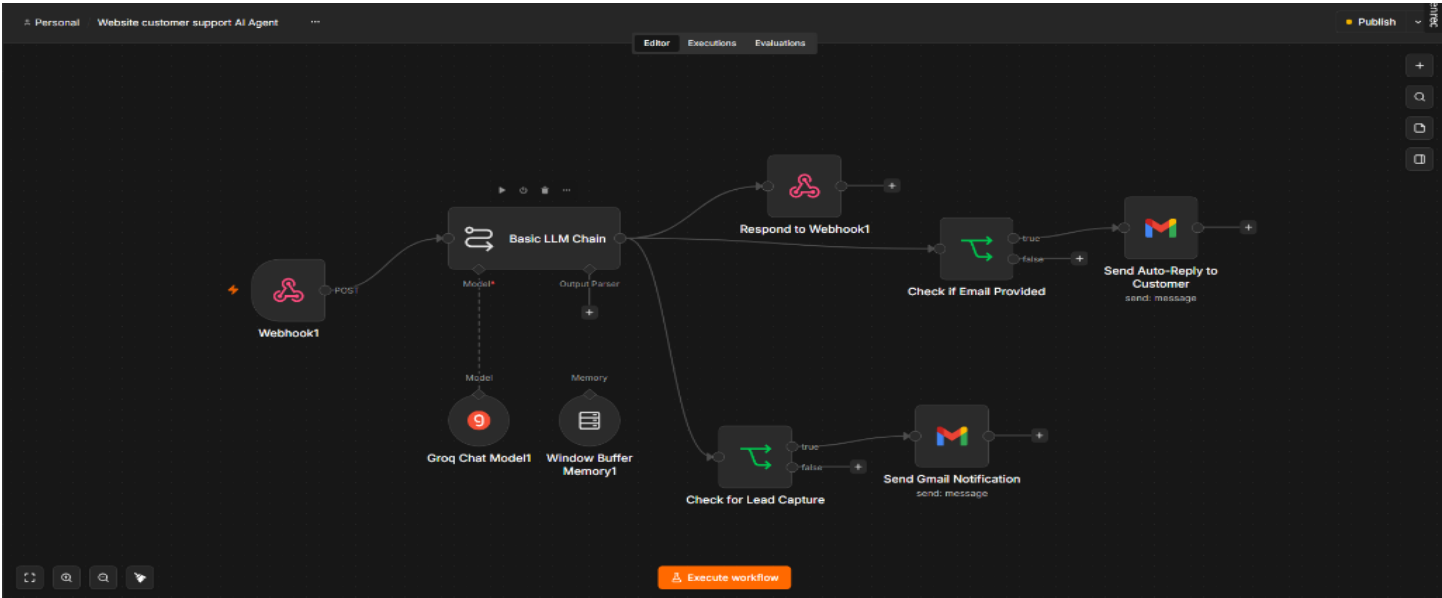
Autonomous Customer Support AI Agent with Memory and Lead Routing

Tech Stack: n8n, LangChain, Groq LLM Engine, Window Buffer Memory, Webhooks, Gmail API

The Challenge and Business Objective

E-commerce and real-estate businesses struggle with 24/7 lead inquiries. Standard rule-based bots fail to understand context, while slow manual responses cause drop-offs. The goal was to build a low-latency AI agent that remembers conversation context, answers queries naturally, captures visitor emails, and immediately dispatches alerts to internal teams.

Production Workflow Architecture Diagram



Engineering Highlights and Deliverables

Sub-Second LLM Ingestion:

Connected ultra-fast Groq Chat Model via LangChain chain inside n8n to respond to incoming webhook chat queries in under 600ms.

Conditional Lead Extraction:

Automated if/else logic detecting email submissions to trigger instant customer welcome messages via Gmail API.

Window Buffer Memory:

Equipped the AI chain with conversational memory, allowing users to ask follow-up questions without repeating context.

Business Impact:

- 24/7 instant response with zero human agent cost.
- 35% increase in captured leads through conversational routing.
- Instant staff alert notifications.

CASE STUDY 03

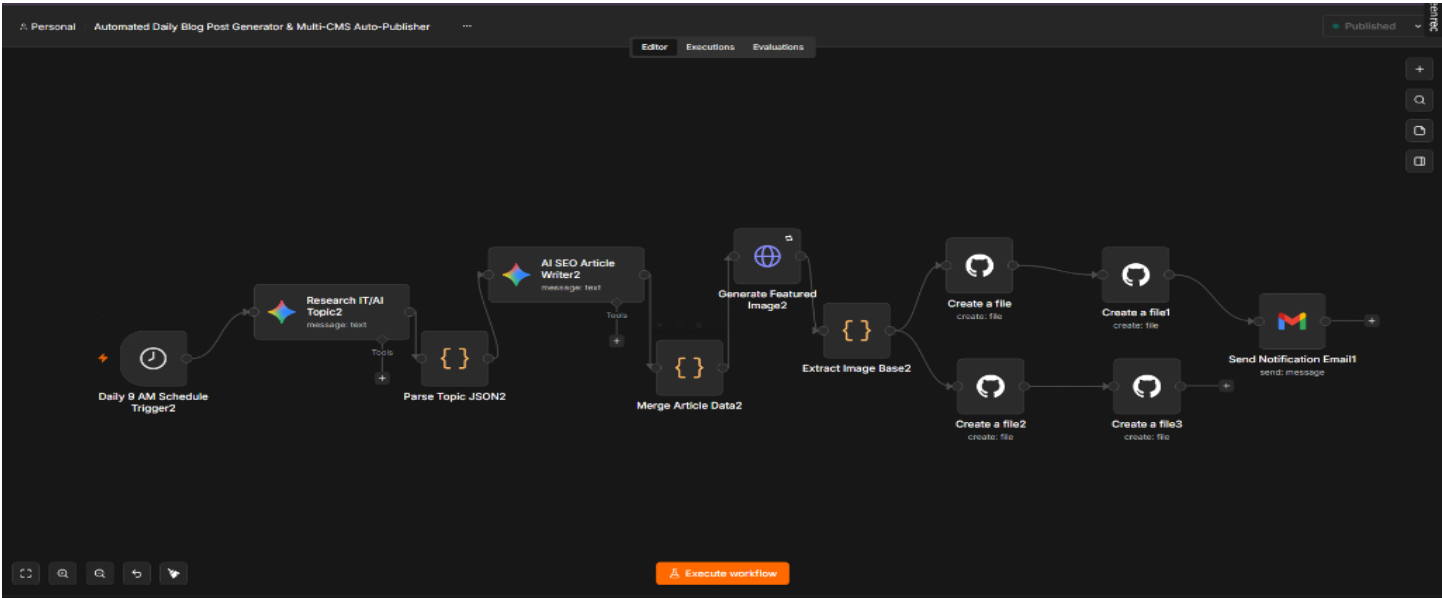
Automated Daily SEO Blog Generator and Multi-CMS Auto-Publisher

Tech Stack: n8n, AI Research Node, SEO Article Writer LLM, DALL-E / Flux API, GitHub API, Gmail

The Challenge and Business Objective

Consistently publishing high-ranking, researched content requires hours of manual topic research, copywriting, featured image creation, markdown formatting, and CMS repository commits. The requirement was a 100% hands-free editorial engine that runs every morning autonomously.

Production Workflow Architecture Diagram



Engineering Highlights and Deliverables

Autonomous Topic Research:

Cron schedule triggers AI research nodes to analyze trending IT/AI topics, structuring output into strict JSON schema.

Automated Git / CMS Deployment:

Commits markdown files and image assets directly to multiple GitHub repository paths for instant static-site deployments.

Dynamic Image Generation & Base64:

Generates custom featured banner art, extracts binary data, and prepares multi-format media assets.

Business Impact:

- Completely hands-free content generation running daily at 9:00 AM.
- Multi-repo GitHub auto-commits with zero downtime.
- Automated email summaries upon publication.